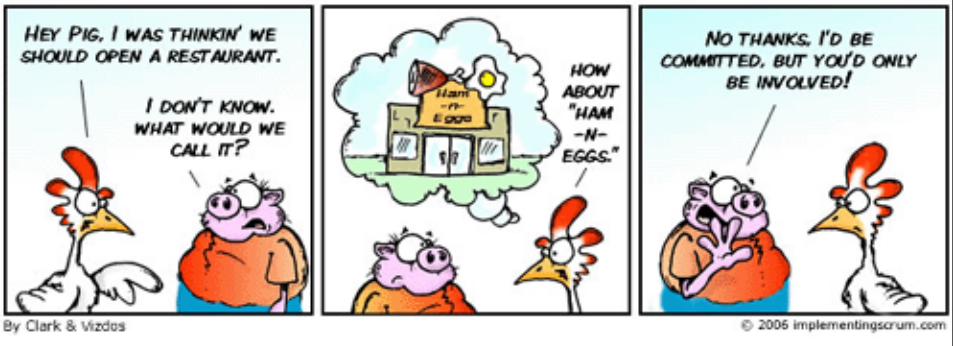
8. Scrum

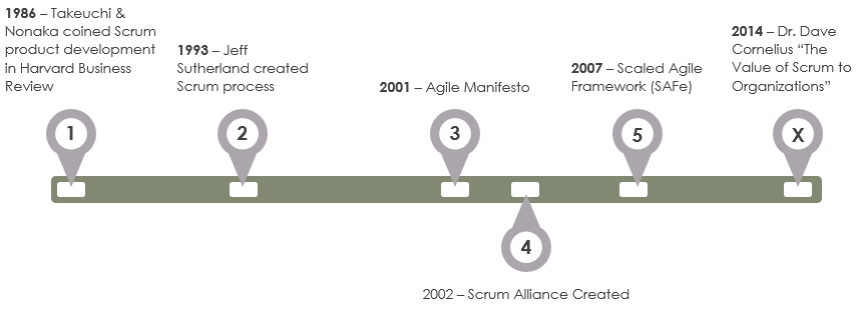
It's about common sense

"Software Development Methodologies"

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)



History of Scrum



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SCRUM

- Scrum is an iterative, incremental method for managing the systems development.
- Emphasizes a set of project management values and practices.
- Does not define any specific software development technique for the implementation phase; it is a wrapper for existing engineering practices.
- Concentrates on how the team members should function, promoting self-directed teams and daily team measurement.
- Produces a potentially shippable set of functionalities at the end of every iteration.
- Has strong emphasis on empirical rather than defined process.

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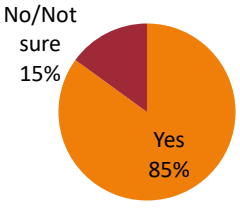
Key practices

- self-directed and self-organizing team
- no external addition of work to an iteration, once chosen
- daily stand-up meeting with special questions
- usually 2 to 4-week iterations
- demo to external stakeholders at end of each iteration
- client-driven adaptive planning

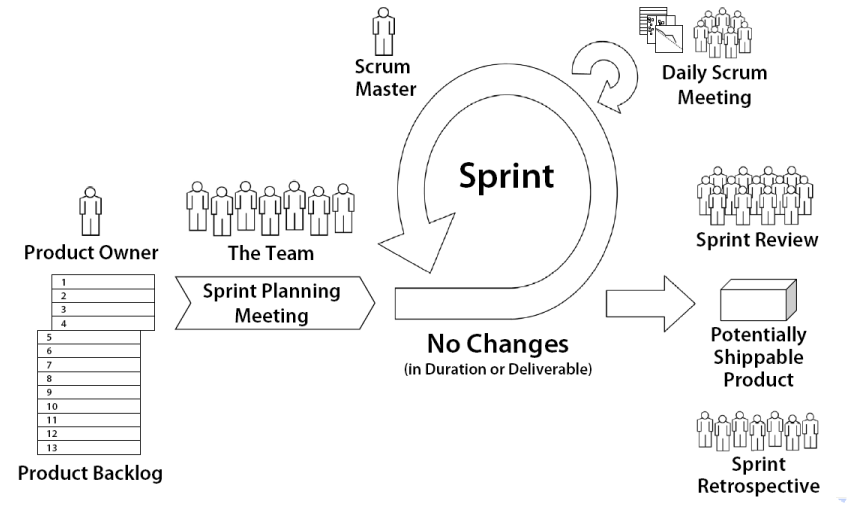
Scrum introduction at Yahoo!

Scrum vs Previous Approach	Worse	Same	Better
Productivity	6%	36%	68%
Morale	9%	39%	52%
Sense of accountability/ownership in project	6%	32%	62%
Collaboration and cooperation in the team	1%	18%	82%
Overall quality of what was developed	10%	46%	44%

- *"If the decision were solely up to you, would your team continue using Scrum?"*



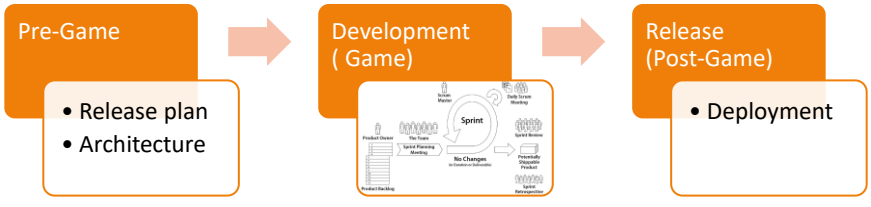
Scrum Process flow



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Scrum Lifecycle



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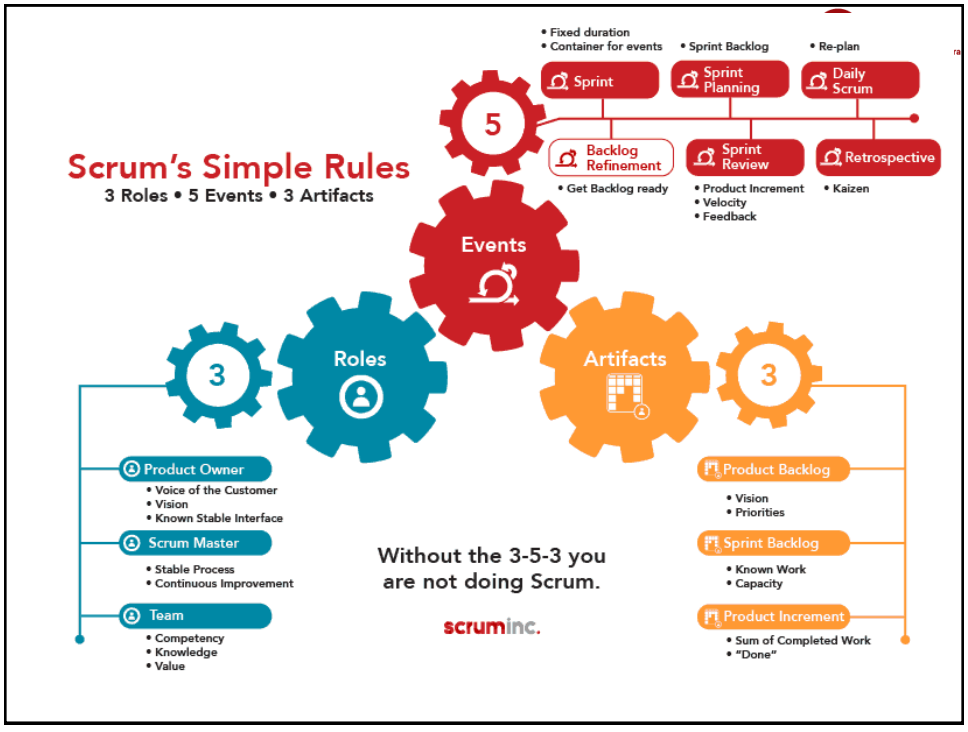
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Lifecycle

1. Pre-game
   1. Planning
      - Write vision, budget, User Stories, Product Backlog with estimates, exploratory design and prototypes
   2. Staging
      - Define requirements for next release, plan next release, exploratory design and prototypes
2. Development
   - Sprint planning, Sprint, Sprint review and retrospective
3. Release (Post-game)
   - Deployment, documentation, training, …

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Scrum framework

- Roles
  - Product Owner
  - Scrum Master
  - Team
- Artifacts
  - Product Backlog
  - Sprint Backlog
  - Product Increment
- Events
  - Sprint
  - Sprint Planning
  - Daily Scrum Meeting
  - Sprint Review
  - Sprint Retrospective

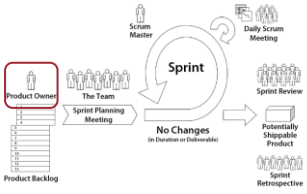
Scrum Roles



Product Owner

- Owns the vision of what should be produced
- Gets inputs from customers, end-users, team, managers, executives, industry experts, etc.
- Is responsible for creating and prioritizing the Product Backlog
- Monitors the project against the stated goals
- Only one person





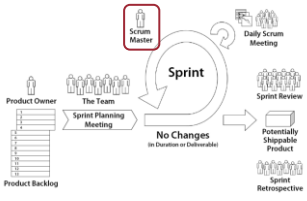
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Scrum Master

- Responsible for setting the team up for success:
  - Establishes Scrum practices and rules
  - Shields the team from outside disturbances
  - Removes obstacles
- Represents management in the project
- Can be played by a team-member.
- Without a Scrum Master, the team has a high risk of failure.





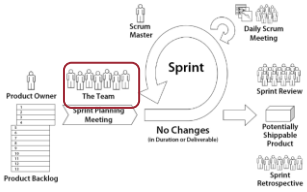
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The Development Team

- Typically 5-9 people
- Cross-functional with undifferentiated roles
  - It has all the skills to produce finished product – designers, coders, testers, etc.
  - Everyone contributes based on competency, rather than just job title
- Self-organizing and self-managing
- Responsible for making a commitment and managing itself to hit the goal
- Authority to do whatever is needed to meet the commitment





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Product Backlog

- Prioritized list of work to be performed on a product
  - Requirements oriented or based on user stories
- Anyone can contribute to backlog items
  - Product Owner is responsible
- Product Owner is responsible for prioritization
  - Focus on business value (80% of the value comes from 20% of the features)
  - Reviewed at the start of each sprint

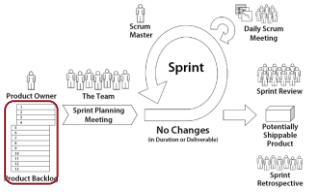


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Product Backlog

	A	B	C	D	E	F
1	**Product Backlog**					
2						
3	**Requirement**	**Num**	**Category**	**Status**	**Pri**	**Estimate**
4	log credit payments to AR	17	feature	underway	5	2
5	process sale-simple cash scenario	232	use case	underway	5	60
6	slow credit payment approval	12	issue	not started	4	10
7	sales commission calculation	43	defect	complete	4	2
8	lay-away plan payments	321	enhance	not started	3	20
9	PDA sale capture	53	technology	not started	1	100
10	process sale-credit pmt scenario	235	use case	underway	5	30



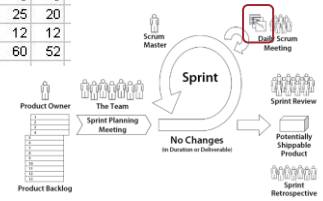
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Sprint Backlog

- Lists all the tasks, and the hours remaining for each
- Updated daily

	A	B	C	D	E	F	G	H	I
1	**Sprint Backlog**								
2	**Task Description**	**Originator**	**Responsible**	**Status**	**Hours of work remaining**				
3					**6**	**7**	**8**	**9**	**10**
4					**362**	**322**	**317**	**317**	**306**
5	Meet to discuss the goals and	JM	JM/SR	Completed	20	10	0	0	0
6	Move Calculations out of	TL	AW	Not Started	8	8	8	8	8
7	Get GEK Data		TN	Completed	12	0	0	0	0
8	Analyse GEK Data - Title		GP	In Progress	24	20	30	25	20
9	Analyse GEK Data - Parcel		TK	Completed	12	12	12	12	12
10	Define & build Database		BR/DS	In Progress	80	80	75	60	52

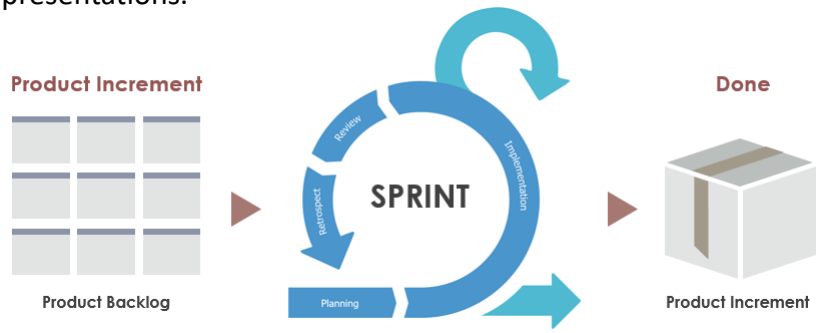


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Product Increment

- The (working) result at the end of each sprint
- The sum of all the product backlog items completed during a sprint plus the value of the increments of all previous sprints
- The result must work (potentially shippable/releasable) –no presentations.

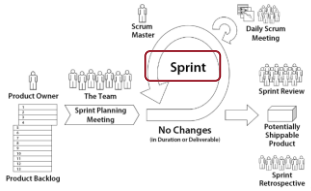


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Sprint

- Fixed Timeframe:
  - Between 2 and 4 weeks.
- Create “done” work
  - Acceptance criteria
- A new Sprint starts after previously one is ended.
- PO might end abruptally the sprint:
  - This must be done only in extreme cases!

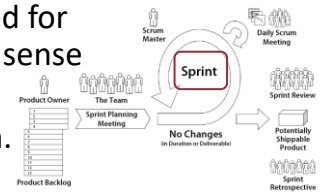


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Sprint

- Fixed period iteration
  - typically between 2- and 4-weeks
- Includes design, coding, testing, and documentation
- During the Sprint, the Scrum Team keeps committed to the goals and end-date
- The Scrum Team is fully responsible for the work
  - Only the Scrum Team can add or remove items in the Sprint backlog
- Abnormal termination of Sprint is called for when the Sprint Goal no longer makes sense
- Sprints occur one after another, without any down-time between them.



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Sprint Planning Meeting

- 2-part meeting to define the next Sprint
- Sprint prioritization
  - Product Owner leads, all Stakeholders are present
  - Analyze and evaluate product backlog
  - Select Sprint goal
  - 4 hours max
- Sprint planning
  - Scrum Team and Product Owner (optional)
  - Decide how to achieve sprint goal (design)
  - Create Sprint Backlog (tasks) from Product Backlog (features/user stories)
  - Estimate Sprint Backlog
  - 4 hours (max 8 hours per 1 month Sprint)

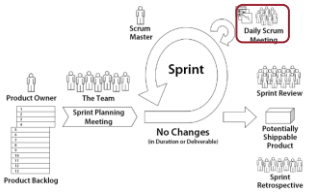


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Daily Scrum Meeting

- Short meeting to update each other on progress and problems
  - Daily, stand up, 15 minutes
  - Usually same place, same time, at the beginning of the day
- Everyone answers 3 questions:
  - What have you done since last daily Scrum?
  - What will you do between now and the next daily Scrum?
  - What is getting in the way (blocks) of meeting the iteration goals?
- Scrum Master notes blocks, and afterwards helps resolve them
  - Not for problem solving
  - Other follow-up meetings can be held
- Others (chickens) can attend the meeting but they do not speak.



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Sprint Review Meeting

- At the end of the Sprint
- Product Owner, Team, Scrum Master, and all Stakeholders
- Demo of what the team has produced.
- The Product Owner gathers feedback from everyone on ways to improve what's been built
  - This feedback is incorporated into the Product Backlog
- Max 4 hours.
- Goal: Inspect product increment.



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Sprint Retrospective Meeting

- Also at the end of the Sprint
- The Team, Product Owner, and Scrum Master
- Goal: review the way of working and look for ways to improve their effectiveness.
- Mechanism for continuous improvement:
  - What went wrong, what went right, what can be done better
  - Which tools, processes, practices can be used to make it better, more productive and enjoyable
- Might be anonymous

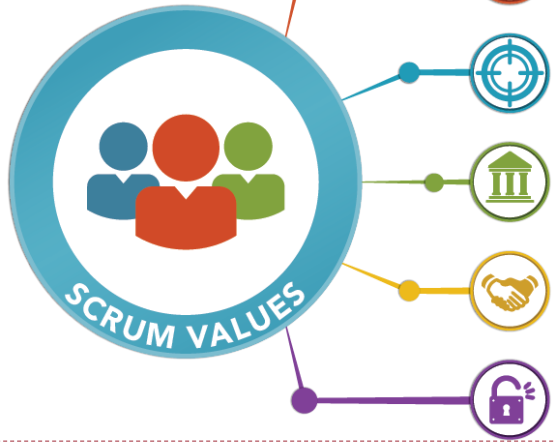


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Scrum Values

Scrum.org



COURAGE
Scrum Team members have courage to do the right thing and work on tough problems

FOCUS
Everyone focuses on the work of the Sprint and the goals of the Scrum Team

COMMITMENT
People personally commit to achieving the goals of the Scrum Team

RESPECT
Scrum Team members respect each other to be capable, independent people

OPENNESS
The Scrum Team and its stakeholders agree to be open about all the work and the challenges with performing the work

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Scrum Values

- Commitment
  - The Scrum Team commits to a defined goal, and decide how best to meet it
  - The Scrum Master commits not to introduce new work, to provide resources, to remove blocks
  - The Product Owner commits to define the Product Backlog and the iteration goal, to provide feedback
- Focus
  - Focus on the goal of the iteration
- Respect
  - Team responsibility rather than scapegoating; the group solves "individual" problems
- Courage
  - Management trusts individuals; team takes responsibility for self-management

Scrum 2020

- Scrum has just celebrated 25 years.
- A new version was launched (Scrum Guide 2020) – January 2021.
- https://scrumguides.org/scrum-guide.html

Scrum 2020

- Smaller framework: From 19 pages (2017) to 13 pages (2021).
- Removed items (daily scrum questions)
- Sprint cancelation section
- …

Scrum 2020

- Development team => Developers
- No more a development “team” inside scrum “team”.
- Only one team: Scrum team.

Scrum 2020

- Introduction of Product Goal
- Introduced the concept of product goal:
  - Give focus for the scrum team
  - Each sprint should gather na incremente towards the product goal.

Scrum 2020

- Introduction of commitments.
- Previous Scrum Guides described Sprint Goal and Definition of Done **without really giving them an identity.**
- Each of the three artifacts now contain 'commitments' to them.

Artifact	Commitments
Product Backlog	Product Goal
Sprint Backlog	Sprint Goal
Increment	Definition of Done

Scrum 2020

- No more self-organizing team => self-managing
- Previous Scrum Guides referred to Development Teams as self-organizing, choosing who and how to do work.
- **Self-organization was widely misinterpreted worldwide** to mean that agile developers do not have to meet commitments.
- **self-managing** will create a more mature attitude towards meeting the Goal of their work

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Scrum 2020

- Introducing of Sprint planning topic “why”:
- Topic One: Why is this Sprint valuable? (2020)
- Topic Two: What can be Done this Sprint?
- Topic Three: How will the chosen work get done?

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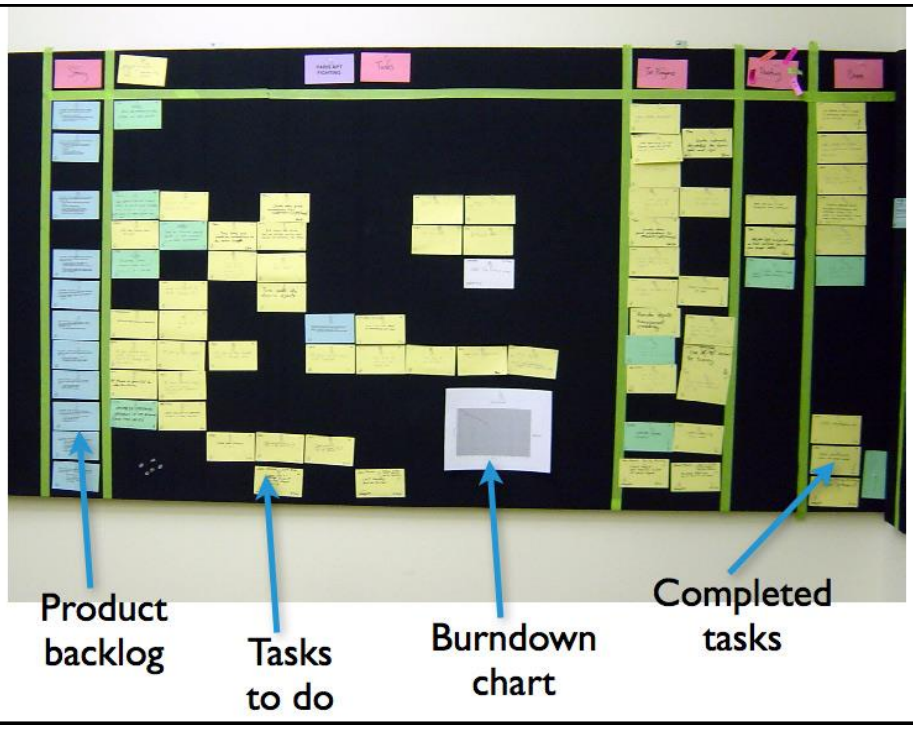
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Scrum 2020

- Simplification of language to a more non-technical audience
  - E.g. : removal of IT words (testing, system, design, etc)
- "Roles" name replaced by "Accountabilities"
- Size of Scrum teams now 10 or fewer (previously was 3 to 9)
- Scrum Master is now a "Leader who serves" not a "Servant Leader"

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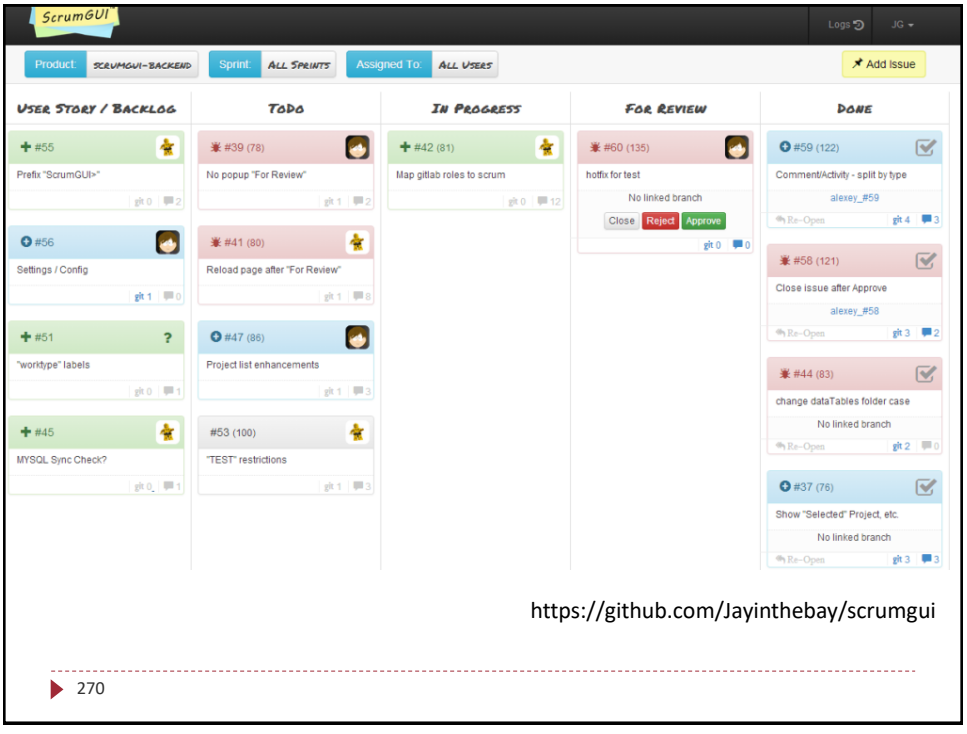
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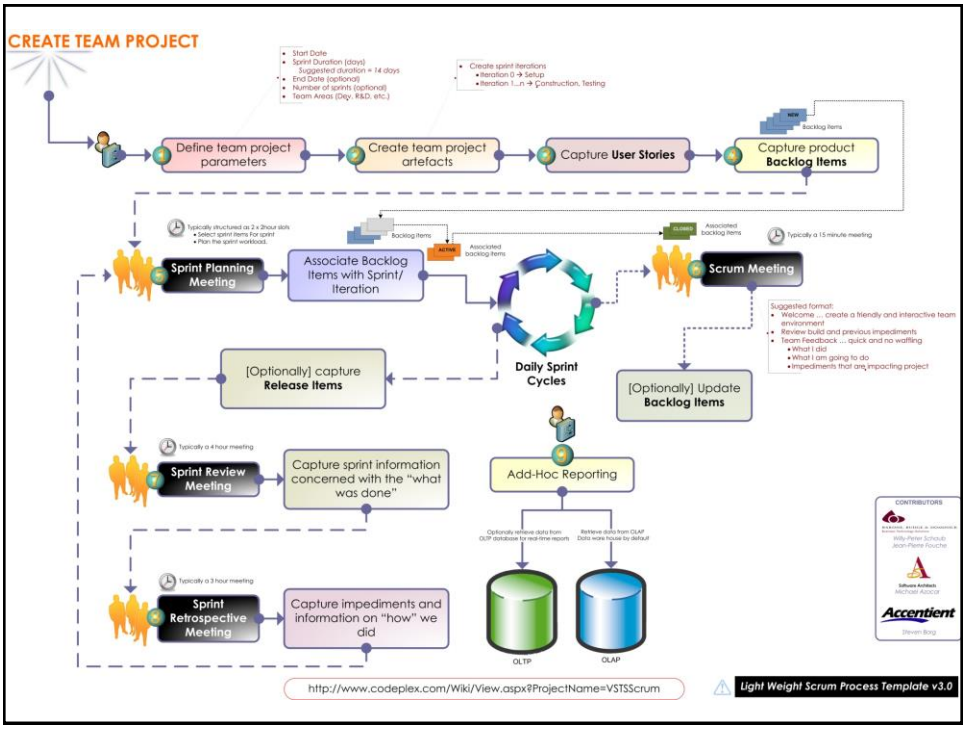
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https://github.com/Jayinthebay/scrumgui

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- http://www.mountaingoatsoftware.com/

- Some figures from the slides were taken from these links

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